Worksheet 2: Thermal Evolution and Differentiation

Covers Lectures 3 and 4: heat sources, convection, core formation, dynamos

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This worksheet is ungraded practice, with the same shape and level as the final exam. Plan up to 2 hours for a serious pass. Work each problem through on paper before you open the solutions, and show your algebra: the exam asks for the steps, not only the number. Some parts state an intermediate result ("show that …"); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps (the worked solutions print five) and round only at the end.

Constants and data

1 yr = 3.156×10^7 s 1 d = 86 400 s $M_\oplus = 5.972 \times 10^{24}$ kg $R_\oplus = 6371$ km $R_{\rm Moon}$ = 1737 km

Thermal diffusivity of rock $\kappa = 10^{-6}$ m^2 s^{-1} Age of the solar system 4.57 Gyr Earth's core mass fraction $x = 0.32$ silicate heat capacity $c_p = 1250$ J kg^{-1} K^{-1}

Isotope	Half-life (Gyr)	Heat production (W per kg isotope)	BSE concentration
^{238}U	4.47	9.5×10^{-5}	20 ppb
^{235}U	0.704	5.7×10^{-4}	0.14 ppb
^{232}Th	14.0	2.6×10^{-5}	80 ppb
^{40}K	1.25	2.9×10^{-5}	28 ppb

Isotope table as in the Lecture 3 notes (BSE = bulk silicate Earth; 1 ppb = 10^{-9} by mass). Hf-W constants as in the Lecture 4 notes: $Q_W = 10^4$, $(^{182}\mathrm{Hf}/^{180}\mathrm{Hf})_0 = 1.02 \times 10^{-4}$, $\lambda = 7.79 \times 10^{-2}$ Myr^{-1} ($t_{1/2} = 8.9$ Myr).

Problem 1 The radiogenic clock

Radioactive decay powers planetary interiors on two very different timescales. Here you compute the long-lived budget from the isotope table, then place both timescales in the planetary context.

(a) The isotope table refers to the bulk silicate Earth (BSE): the mantle plus crust, without the core. Compute the BSE mass from $M_\oplus$ and the core mass fraction x (show that $M_{\rm BSE} \approx 4.06 \times 10^{24}$ kg), then compute the present-day specific heating rate of BSE rock by summing the four isotopes (show that it is 4.872×10^{-12} W kg^{-1}). Using the round value $M_{\rm BSE} = 4.0 \times 10^{24}$ kg, give Earth's present radiogenic power in TW and compare it with the value quoted in the notes.

(b) The radiogenic-heating figure in the Lecture 3 notes shows the short-lived isotope ^{26}Al ($t_{1/2} = 0.72$ Myr) crossing below the summed long-lived curve. Read off the approximate crossover time from the figure. How many ^{26}Al half-lives is that (show that it is ≈ 14), and by what factor has ^{26}Al decayed by then?

(c) In 2 to 3 sentences: why do U, Th, and K reside in the mantle and crust rather than in the core, and what does this imply for *where* radiogenic heat is released inside Earth?

Problem 2 Conduction fails, convection takes over

Whether a body can cool by conduction alone is a question of size. Here you find the size limit, then show that Earth's mantle convects instead, and how vigorously.

(a) Compute the conductive cooling timescale $\tau = L^2/\kappa$ for the Moon, using $L = R_{\text{Moon}}$ (show that $\tau \approx 9.6 \times 10^{10}$ yr), and compare with the age of the solar system. Then invert the timescale: what is the largest body size that *can* cool conductively within 4.57 Gyr?

(b) Compute the Rayleigh number of Earth's mantle,

$$\text{Ra} = \frac{\alpha\, \rho\, g\, \Delta T\, d^3}{\kappa\, \eta},$$

with $\alpha = 2 \times 10^{-5}\ \text{K}^{-1}$, $\rho = 4000\ \text{kg}\,\text{m}^{-3}$, $g = 10\ \text{m}\,\text{s}^{-2}$, $\Delta T = 2500$ K, mantle depth $d = 3000$ km, and dynamic viscosity $\eta = 10^{21}$ Pa s (show that $\text{Ra} \approx 5.4 \times 10^7$). Compare with the critical value $\text{Ra}_c \approx 1708$.

(c) Judge this claim in 2 to 3 sentences: *"Doubling the mantle's temperature contrast ΔT would double the convective heat flux."*

Problem 3 A minimal thermal evolution model

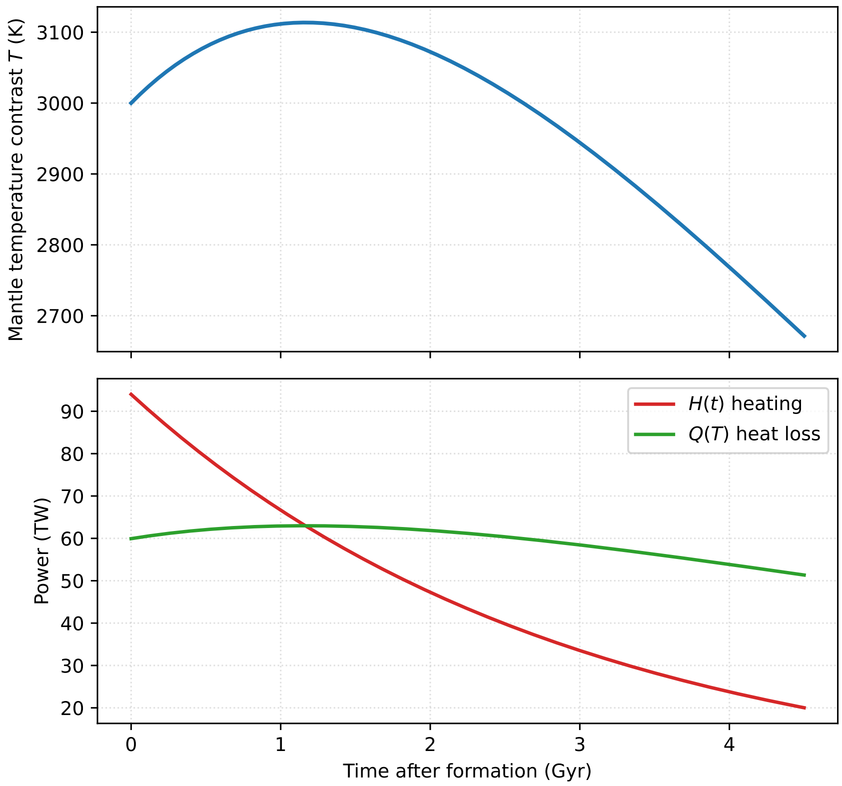
The heat budget of the Lecture 3 notes (radiogenic heating in, convective heat loss out) can be turned into a one-line model of Earth's thermal history. This problem assembles the model on paper and then reads Earth's thermal history from its numerical solution.

(a) Problem 2(c) already used the scaling $q \propto \Delta T^{4/3}$; derive it now in one line from $\text{Nu} \propto \text{Ra}^{1/3}$ and $\text{Ra} \propto \Delta T$ (all other parameters fixed). Writing T for the mantle temperature contrast, energy conservation for the silicate shell then reads

$$C\,\frac{\mathrm{d}T}{\mathrm{d}t} = H(t) - Q(T), \qquad Q(T) = Q_0\left(\frac{T}{T_0}\right)^{4/3}, \qquad H(t) = H_\mathrm{f}\, e^{-t/\tau_r},$$

with $C = M_{\text{BSE}}\, c_p$ the heat capacity of the silicate shell. State in one sentence each what the two right-hand terms represent.

(b) The figure shows the numerical solution of the model, integrated from $T(0) = 3000$ K over 4.5 Gyr with $H_\mathrm{f} = 94$ TW, $\tau_r = 2.91$ Gyr, $Q_0 = 47$ TW at $T_0 = 2500$ K, and $C = 4.0 \times 10^{24} \times 1250 = 5.0 \times 10^{27}\ \text{J}\,\text{K}^{-1}$.



Read off: the maximum temperature and when it occurs, the final temperature $T(4.5\text{ Gyr})$, and the final Urey ratio H/Q. Then explain in one or two sentences why the temperature first *rises* although the heat source only decays.

(c) Set $H = 0$ in the model and solve it analytically: show that at late times the temperature falls off as $T \propto t^{-3}$. What does this limit say about how quickly a small, radiogenically dead body fades?

Problem 4 Building a core and dating it

Core formation is fast, chemically selective, and datable. Here you predict what it leaves behind in the mantle, and read the isotopic clock it starts.

(a) Let x be the core mass fraction and $D = C^{\text{metal}}/C^{\text{silicate}}$ the metal-silicate partition coefficient of a trace element. Assuming the whole planet equilibrated once, show from mass conservation that the mantle retains a fraction

$$\frac{C^{\text{silicate}}}{C^{\text{bulk}}} = \frac{1}{1 + x\,(D-1)}$$

of the bulk concentration. Evaluate it for iridium with $D = 10^4$ and $x = 0.32$ (show that the fraction is $\approx 3.1 \times 10^{-4}$).

(b) The observed iridium depletion of Earth's mantle is a factor ~ 1000, i.e. the mantle retains ~ 10^{-3} of chondritic. Compare with your prediction from part (a) and explain, in 2 to 3 sentences, what the discrepancy implies. What single observation about the element *ratios* clinches the interpretation?

(c) A planetesimal's mantle shows a tungsten excess $\varepsilon^{182}\text{W} = +20$ with an Hf/W enrichment factor $(\text{Hf}/\text{W})/(\text{Hf}/\text{W})_{\text{CHUR}} = 26$, so the bracketed factor in the notes' two-stage equation is 25. Using the Hf-W constants from the data box, show that $e^{-\lambda t_c} = 0.784$ and find the two-stage core-formation age t_c. Compare with the accretion-timeline figure in the Lecture 4 notes.

(d) In 2 to 3 sentences: why can the Hf-W chronometer not date any core-forming event later than roughly 45 Myr after CAIs, and what tool takes over for such late events?

Problem 5 Dynamos and shields

A convecting metallic core can maintain a magnetic field, and that field carves a cavity in the solar wind. Here you test both, for two very different bodies.

(a) Start from pressure balance at the dayside magnetopause, $\frac{1}{2}\rho_{sw}v_{sw}^2 = B_{mp}^2/(2\mu_0)$, with a dipole field $B(r) = B_0\,(R_p/r)^3$. Show that the standoff distance obeys

$$r_{mp} = R_p\left(\frac{B_0^2}{\mu_0\,\rho_{sw}v_{sw}^2}\right)^{1/6}.$$

During a coronal mass ejection the solar-wind density rises tenfold at fixed speed. Starting from Earth's typical $r_{mp} = 10\,R_\oplus$, show that the magnetopause moves in to $\approx 6.8\,R_\oplus$.

(b) Ganymede's dynamo operates in a liquid iron shell of thickness $L \approx 300$ km with magnetic diffusivity $\eta = 1\ \text{m}^2\ \text{s}^{-1}$. If its convection stopped, the field would decay ohmically on the timescale $\tau_{ohm} = L^2/\eta \approx 3 \times 10^3$ yr (verify this in one line if you wish). Explain in 2 to 3 sentences what this timescale, set against the age of the solar system, proves about the field the Galileo spacecraft measured in 1996, and why a "fossil field" interpretation fails.

(c) Judge this claim in 2 to 3 sentences: *"Venus has no magnetic field because it rotates too slowly."*